Abstract Algebra

Summary note of Dummit & Foote's Abstract Algebra

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Contents

Chapter 1

Introduction to Groups

1.1 Groups

Binary operation

1. A *binary operation* $\star$ on a set G is a map $\star : G \times G \to G$. We denote $\forall a, b \in G, \star(a, b)$ by $a \star b$.

2. A binary operation $\star$ on a set G is *associative* if

$$\forall a, b, c \in G, a \star (b \star c) = (a \star b) \star c$$

3. A binary operation $\star$ on a set G is *commutative* if

$$\forall a, b \in G, a \star b = b \star a$$

4. For a set G, $H \subseteq G$ and a binary operation $\star$ on G, if the restriction of $\star$ to H is a binary operation on H, H is said to be *closed* under $\star$.

Group

1. A *group* is an ordered pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G, which satisfies

   (a) $\star$ is associative.

   (b) $\exists\ e \in G, \forall a \in G, a = a \star e = e \star a$

   (c) For e of (b), $\forall a \in G, \exists a^{-1} \in G, a \star a^{-1} = a^{-1} \star a = e$

2. Group $(G, \star)$ is called *abelian* when $\star$ is also commutative.

Direct product

Let $(A, \star)$ and $(B, \diamond)$ are groups. We can form group $A \times B$ called *direct product* of A and B by

$$A \times B = \{(a, b) | a \in A, b \in B\}$$

with following binary operation $\cdot$:

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 \star a_2, b_1 \diamond b_2)$$

Here, one can claim that $(A \times B, \cdot)$ is a group. The associativity of $\cdot$ is clear. And let's denote the identity of $(A, \star), (B, \diamond)$ by e_1, e_2 respectively. We can easily check that (e_1, e_2) is identity of $(A \times B, \cdot)$ and $(e_1, e_2) \in A \times B$. This holds group axiom (2). Finally, $\forall a \in A, b \in B$, we can compose (a^{-1}, b^{-1}). Then by the definition of $\cdot$, $(a, b) \cdot (a^{-1}, b^{-1}) = (a^{-1}, b^{-1}) = (e_1, e_2)$ holds. So, $(a, b)^{-1} = (a^{-1}, b^{-1})$, holds group axiom (3).

Propositions on Groups

Let $(G, \star)$ be any group. Following always holds.

1. The identity of G is unique.
2. $\forall a \in G$, a^{-1} is uniquely determined.
3. $\forall a \in G, (a^{-1})^{-1} = a$
4. $\forall a, b \in G, (a \star b)^{-1} = b^{-1} \star a^{-1}$
5. $\forall a_1, \cdots, a_n \in G$ the value of $a_1 \star a_2 \star \cdots \star a_n$ is independent of how the expression is bracketed.

All the proofs are easy.

Cancellation Laws hold in G

1. $au = av \implies u = v$
2. $ub = vb \implies u = v$

Group order

For any group G and $x \in G$, let's say the *order* of x (denote by $|x|$) by

$$|x| = \underset{n \in \mathbb{N}}{\operatorname{argmin}} \{x^n = 1\}$$

If such n does not exists, $|x| = \infty$.

1.2 Dihedral Groups

The definition of *dihedral group* is quite abstracted.

Dihedral Groups

One natural, and important structure is symmetries of geometric objects. For $n \in \mathbb{Z}^+, n \geq 3$, the *dihedral group* D_{2n} be the set of symmetries on regular n-gon, where a symmetry is any rigid motion.

Now let's imagine that we label each vertices of such n-gon as $\{1, 2, \cdots, n\}$. Then, each symmetries can be viewed a permutation σ on $\{1, 2, \cdots, n\}$. Then now we can formulate the group D_{2n} with composition $\circ$. Let $s, t \in D_{2n}$ be two symmetries. Composition $s \circ t = st$, is defined by apply t and then s.

One interesting thing is that, $|D_{2n}| = 2n$. It can be proved. In every symmetry, vertex 1 and 2 are adjacent. So, our job is clear. Choose the position of vertex 1. (n cases) And then, choose the position of vertex 2 which adjacent with vertex 1. (2 cases) So, we have $2n$ cases total, implies that $|D_{2n}| = 2n$. Or, we can enumerate what such $2n$ symmetries are. n symmetries are came from rotation by $2\pi i/n, i \in \{0, \cdots, n-1\}$. Remaining n symmetries are came from flip(reflection), with n axis of n-gon.

From now on, we'll fix our setting to understand D_{2n} further. Fix a regular n-gon centered at the origin on $\mathbb{R}^2$ and label the vertices in clockwise manner, from 1 to n. And denote $r \in D_{2n}$ be a $\frac{2\pi}{n}$ rotation. And denote $s \in D_{2n}$ be a reflection via axis containing vertex 1. We can observe following propositions.

Propositions on D_{2n}

1. $1, r, r^2, \cdots, r^{n-1}$ are distinct, and $|r| = n$
2. $|s| = 2$
3. $s \neq r^i$ for any $i \in \{0, 1, \cdots, n-1\}$
4. $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$. Consequence of this is that,
$$D_{2n} = \{1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}\}$$
5. $rs = sr^{-1}$, So D_{2n} is *non-abelian*
6. $r^i s = sr^{-i}, 0 \leq i \leq n$

Everythings are not hard to prove.

Now, let's discussion about *generators* and *relations*. As we can see in above study, to represent all elements of D_{2n}, two element r, s are sufficient. Similarly, we introduce the notions of *generators* and *relations* for arbitrary groups. Although we recall both notions later, let's bring the notions of this basis-like things.

Generator of G

For a group G, $S \subset G$ is called a set of *generators* if every element of G can be written as a (finite) product of elements of S and their inverses. Also we write $G = \langle S \rangle$ and say that G is *generated by* S.

For clear understanding, $\mathbb{Z} = \langle 1 \rangle, D_{2n} = \langle r, s \rangle$.

Relation in G

Any equations in group G that the generators of G satisfy are called *relations* in G.

For example, we have 3 relations in D_{2n} :

$$r^n = 1, s^2 = 1, rs = sr^{-1}$$

And any other relation between elements of D_{2n} can be derived from these three. (later) By the way, this notion enable us to represent group in simple way. (Only few elements and equations) This leads us to following definition.

Presentation of G

Let's imagine that a group G is generated by $S \subseteq G$ and there are some collection of relations $R_1, \cdots, R_m$ such that any relation among the S is consequence of them. In this case, we write

$$G = \langle S | R_1, R_2, \cdots, R_m \rangle$$

and call this by *presentation* of G.

In this step, this notion is quite verbose. So, we'll discuss it later of this book.

1.3 Symmetric Groups

Now let's move on to next step. Actually our goal is abstraction of natural notions. In this section, we'll derive set of all bijections(or permutations).

Symmetric Group on Ω

Let Ω be any set. Then define S_Ω by the set of all bijections from Ω to Ω. We can also understand S_Ω is set of all permutations of Ω. For the binary operation $\circ$ which is function composition, $(S_\Omega, \circ)$ is a group. (Check!) We call $(S_\Omega, \circ)$ by *symmetric group* on Ω.

One natural consideration is that, when $\Omega = \{1, 2, \cdots, n\}$. In this case, we denote the symmetric group by $(S_n, \circ)$ and call it *symmetric group of degree* n. We already know that $|S_n| = n!$

Let's move on. Can we find some interesting observations from each $\sigma \in S_n$?

Cycle decomposition

A *cycle* is a string of integers represents the element of S_n which cyclically permutes these integers and fixed all others. For example, a cycle $(a_1, \cdots, a_m) \in S_n$ is the permutation such that sends $a_1 \mapsto a_2, \cdots, a_{m-1} \mapsto a_m, a_m \mapsto a_1$
In general, we can decompose for each $\sigma \in S_n$ into k cycles, means that

$$\forall \sigma \in S_n, \sigma = (a_1\ a_2 \cdots a_{m_1})(a_{m_1+1} \cdots a_{m_2}) \cdots (a_{m_{k-1}+1} \cdots a_{m_k})$$

With this, we can easily compute $\sigma(x)$ for any $x \in [n]$. Find the cycle that x is contained, then the immediately right-next element is $\sigma(x)$. This decomposition is called *cycle decomposition of* σ

We can easily obtain the cycle decompositioning algorithm, so skip. And with this, we can easily compute cycle decomposition of σ^{-1}. Just write the numbers in each cycle of the cycle decomposition of σ in reverse order. i.e.,

$$\sigma^{-1} = (a_{m_1}\ a_{m_1-1} \cdots, a_1)(a_{m_2}\ \cdots a_{m_1+1}) \cdots (a_{m_k} \cdots a_{m_{k-1}+1})$$

Also, we can observe that $\forall n \geq 3, S_n$ is not abelian group. This is because that $(1\ 3) \circ (1\ 2) \neq (1\ 2) \circ (1\ 3)$. And note that, we can view σ by composition of each cycles. (since cycles are permutations of S_n itself) Moreover, they are independent to order. i.e., *disjoint cycles commute.* And also, each cycles are independent to cyclic permutation of inner elements. i.e., $(1\ 2\ 3) = (2\ 3\ 1) = (3\ 1\ 2)$

Proposition 1.3.1.

The order of $\sigma \in S_n$ equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Proof. $\square$

1.4 Matrix Groups

Actually, field is our later interests. However, to understand group theory, very powerful tool is understanding linear algebra. So we alert matrix groups, whose coefficients come from fields. For sake of goal, let's describe what the fields are.

Fields

A *field* is a set F with two **commutative** binary operations $+, \cdot$ on F such that $(F, +)$ forms an abelian group with identity 0, and $(F \setminus \{0\}, \cdot)$ also forms an abelian group with identity 1, and following *distributive law* holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c), \forall a, b, c \in F$$

And we denote $F^\times = F \setminus \{0\}$

Actually our interests are groups of matrices over F. Let's define following important group here.

Matrix groups($GL_n(F)$)

$$GL_n(F) = \{A | A \in Mat_{n\times n}(F), \det(A) \neq 0, \text{i.e., invertible}\}$$

Let binary operation $\cdot$ denote matrix multiplication. It is easy to see that $(GL_n(F), \cdot)$ is a group. It called *general linear group of degree n*.

1.5 Quaternion Group

In mathematics, there are some famous counter-examples when we conject some properties on mathematical structures. This group is sort of them.

Quaternion Group

The *quaternion group* Q_8 is defined by

$$Q_8 := \{1, -1, i, -i, j, -j, k, -k\}$$

Here, we define binary operation $\cdot$ as follows;

$$\forall q \in Q_8, 1 \cdot q = q \cdot 1 = q$$

$$(-1) \cdot (-1) = 1, \quad \forall q \in Q_8, (-1) \cdot q = q \cdot (-1) = -q$$

$$i \cdot i = j \cdot j = k \cdot k = -1$$

$$i \cdot j = k \quad j \cdot i = -k$$

$$j \cdot k = i \quad k \cdot j = -i$$

$$k \cdot i = j \quad i \cdot k = -j$$

(I'll add more intuition about this group after I understand well.)

1.6 Homomorphisms and Isomorphisms

By defining groups, we abstracted many mathematical structures into group theories. Then now, we can imagine the maps between them and what the 'look the same group' is. First, let's define the structure-preserved maps on groups.

Homomorphism

Let $(G, \star), (H, \diamond)$ be groups. A map $\varphi : G \to H$ such that

$$\forall x, y \in G, \varphi(x \star y) = \varphi(x) \diamond \varphi(y)$$

is called *homomorphism* from G to H.

Then now, we define more strong (and more precise to our goal) concepts: Two groups are conceptually equal.

Isomorphism

Let the map $\varphi : G \to H$ be an homomorphism from G to H. We call that φ is an *isomorphism* when φ is a bijection. If such isomorphism exists, then we denote $G \cong H$ and say that G and H are *isomorphic*.

On the high level, $G \cong H$ if there exists bijective structure-preserve map φ. And if $G \cong H$, they are exactly same group except that how we write the elements and operation.

One of the most familar example is that $(\mathbb{R}, +) \cong (\mathbb{R}^+, \times)$. Such isomorphism is explicitly given as $\varphi(x) = e^x$. It is bijective since $\varphi^{-1}(a) = \log a$. And clearly homomorphism, since $e^{x+y} = e^x e^y$. So, both are exactly same structure.

Following fact is quite not obvious:

First isomorphim classes

Any non-abelian group of order 6 is isomorphic to S_3. Further, every group whose order is 6 is isomorphic to one of two groups: $S_3, \mathbb{Z}/6\mathbb{Z}$ and so, $S_3 \ncong \mathbb{Z}/6\mathbb{Z}$.

One of main goal in group theory is that learn skills to prove this kind of statements.